7.8 Security Threats and Vulnerabilities

As drones become more accessible, intelligent, and widespread, they also present increasing security threats and vulnerabilities. Originally developed for defense and industrial applications, drones are now used across various civilian sectors – ranging from logistics and infrastructure inspection to photography and law enforcement. However, this accessibility also enables malicious actors to exploit drone technology for unlawful or harmful purposes. Concerns include their potential misuse in criminal activities, cybersecurity breaches, and threats to national and public safety.

The challenge lies in balancing the advantages of drone innovation with effective risk management and defense mechanisms. With autonomous capabilities, long-range control, and the ability to carry payloads, drones can be misused in ways that are difficult to detect or prevent using traditional security frameworks. Addressing these risks requires a combination of technological countermeasures, legal enforcement, and international cooperation to safeguard both airspace integrity and public security.

7.8.1 Potential misuse of drones for illegal activities

The proliferation of affordable, high-performance drones has led to a growing risk of their misuse for criminal or terrorist purposes. Drones can be easily modified to perform illicit actions such as smuggling, espionage, unauthorized surveillance, and even weaponization. Their small size, maneuverability, and ability to operate remotely make them difficult to detect, especially in urban or restricted areas.

Common illegal activities involving drones include smuggling contraband into prisons or across borders, spying on sensitive installations, and unauthorized aerial photography or data collection. More concerningly,